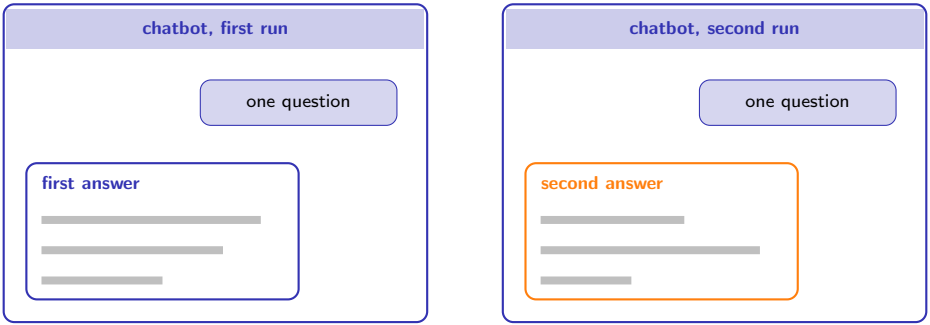
Picking the Next Token, Greedy Decoding to Beam Search

Text Analytics, Chapter 7

Prof. Dr. Jörg Osterrieder

One Question Sent Twice

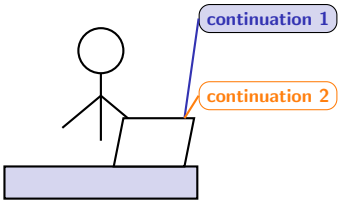


One question sent twice to a chatbot can have two different answers.

One Prompt, Two Continuations

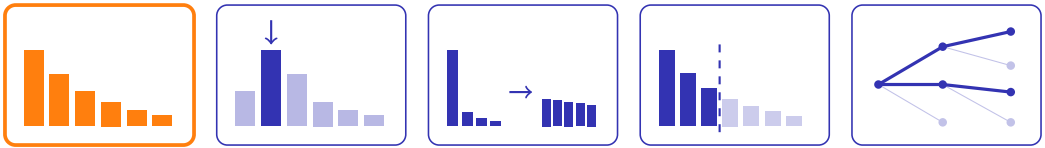
Scenario: You send one prompt to a chatbot twice and receive two different continuations.

- After the prompt, the language model has a probability p_i for each token i of its vocabulary V
- One token is appended, and the step is repeated on the longer text
- With a random draw at each step, the texts of two runs differ
- The question: which token is appended, under which rule



The appended token depends on the decoding rule.

One Distribution, Five Decoding Rules



▶ **1 Next-Token Distribution** | 2 Greedy Decoding | 3 Temperature | 4 Top-k, Top-p | 5 Beam Search

▶ next

One next-token distribution is the input of all five decoding rules.

Learning Goals for Decoding

▶ **1 Next-Token Distribution**
Compute next-token probabilities from scores; write the probability of a continuation as a product.

2 Greedy Decoding
Name the greedy token; explain the repetition loop.

3 Temperature
Name the limits of $p^{(T)}$ as $T \to 0$, $T \to \infty$.

4 Top-k, Top-p
Choose top-k or top-p for a peaked or a flat distribution.

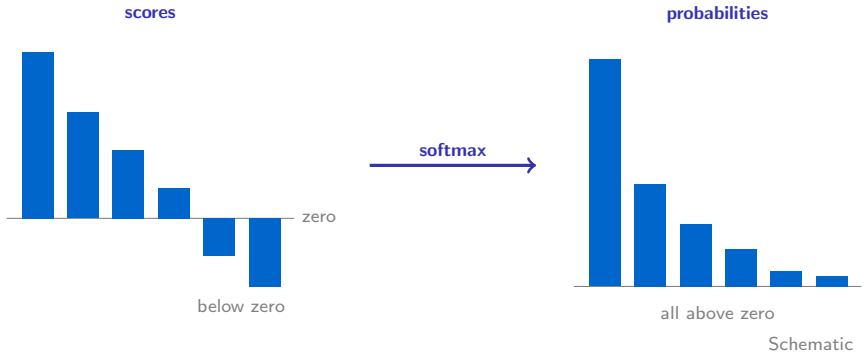
5 Beam Search
Run beam search of width B on a two-step tree.

6 Worked Example: GPT-2 Small
Compute each rule on the numbers of GPT-2 small after one prompt.

▶ next under each part ahead: its learning goal

Worked Example: GPT-2 small after the first words of *Pride and Prejudice*.

From Scores to Probabilities



A larger score gives a larger probability, and every probability is above zero.

Softmax Probabilities From Scores

Definition

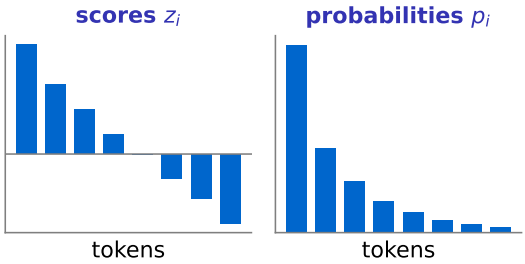
After a context c, the language model has a **score** z_i for each token i of the vocabulary V. The **softmax** of the scores is the next-token probability

$$p_i = \frac{e^{z_i}}{\sum_{j \in V} e^{z_j}}.$$

Intuition:

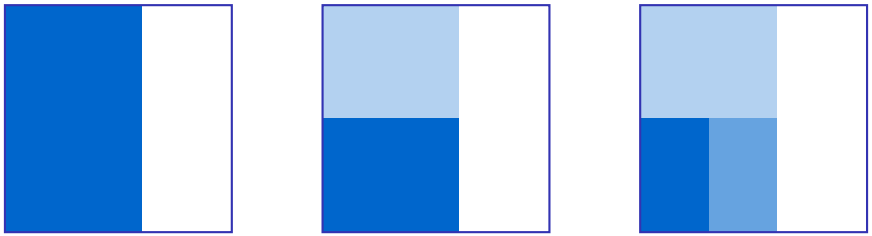
- A larger score $\rightarrow$ a larger probability
- $p_i > 0$ for every token, $\sum_i p_i = 1$
- $z_i + a$ for every token $\rightarrow$ the same p: p depends only on differences of scores

Schematic: softmax of the scores



The softmax of the scores is the next-token distribution p.

A Longer Continuation Has a Smaller Probability



With each token, P is multiplied by a fraction.

Probability of a Continuation as a Product

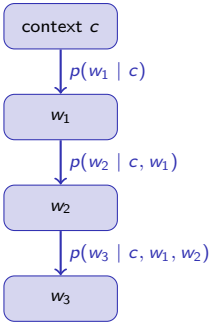
Definition

A **continuation** $w_1, \ldots, w_n$ of the context c has the probability

$$P = \prod_{t=1}^{n} p(w_t \mid c, w_{<t}), \quad \log P = \sum_{t=1}^{n} \log p(w_t \mid c, w_{<t}).$$

Intuition:

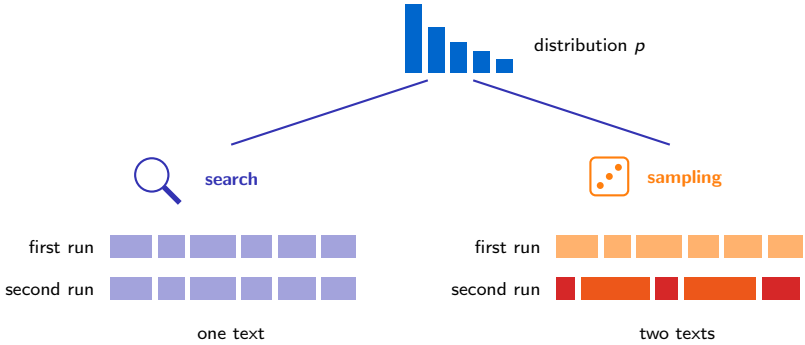
- Each factor is at most $1 \rightarrow P$ is smaller with every token
- $\log P \leq 0$; closer to $0 \rightarrow$ a more probable continuation
- A large first factor can be followed by small ones



$$P = p(w_1 \mid c)\, p(w_2 \mid c, w_1)\, p(w_3 \mid c, w_1, w_2)$$

The probability of a continuation is the product of its next-token probabilities.

Search, Sampling



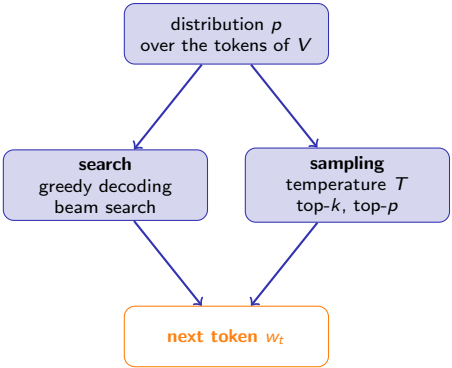
Under a decoding rule, the next token is found by a search or drawn at random.

Decoding Rules: Search, Sampling

Definition

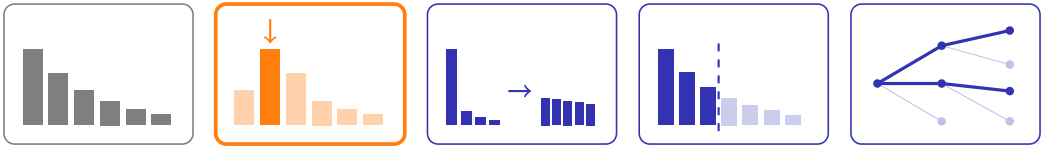
A **decoding rule** is a map from the distribution p at a step to the next token: by a **search** for a probable continuation, or by a **draw** from a distribution q derived from p.

- Search: greedy decoding, beam search → one text for one prompt
- Sampling: temperature T, top-k, top-p → a new text on each run
- Temperature first, then a cut by top-k or top-p: the sampling rules are combined



Under search, one prompt gives one text; under sampling, each run gives a new text.

One Distribution, Five Decoding Rules



✓ finished ▶ next

One next-token distribution is the input of all five decoding rules.

Greedy Decoding

✓ 1 Next-Token Distribution

▶ **2 Greedy Decoding**
Name the greedy token; explain the repetition loop.

3 Temperature
Name the limits of $p^{(T)}$ as $T \to 0$, $T \to \infty$.

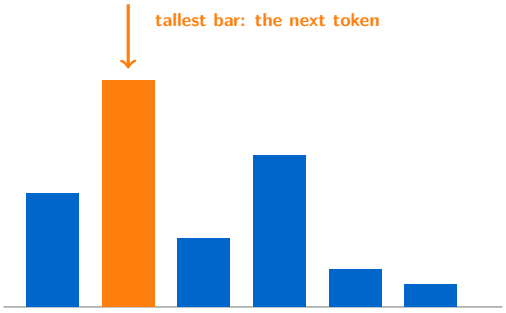
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Compute each rule on the numbers of GPT-2 small after one prompt.

✓ finished ▶ next under each part ahead: its learning goal

Greedy Decoding: the Tallest Bar



Schematic

Under greedy decoding, there is no draw: the token of the tallest bar is appended.

The Greedy Token Has the Largest Probability

Definition

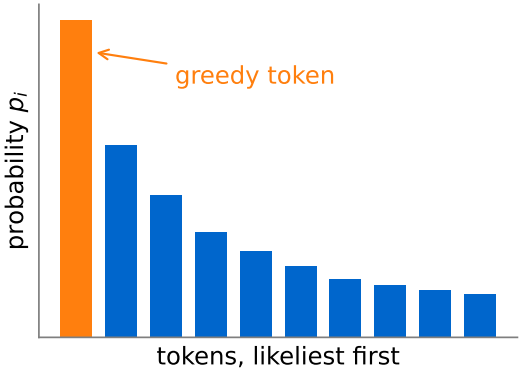
Under **greedy decoding**, the next token is the token with the largest probability, at every step:

$$w_t = \arg\max_{i \in V} p(i \mid c, w_{<t}).$$

Intuition:

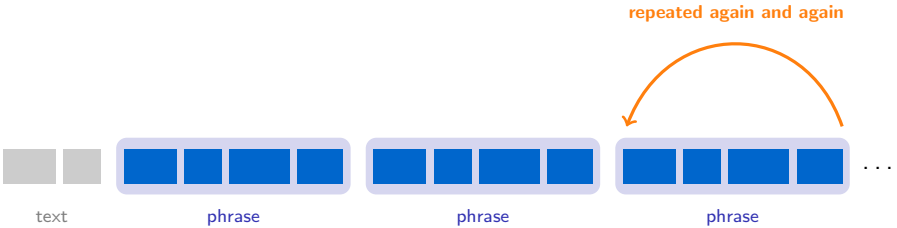
- No draw $\rightarrow$ one prompt, one text, on every run
- Each step is decided alone: later probabilities are left out of the comparison
- Greedy decoding is temperature at $T \rightarrow 0$ and beam search at $B = 1$

Schematic: greedy token in orange



Under greedy decoding, the likeliest token is appended at every step.

One Phrase Repeated



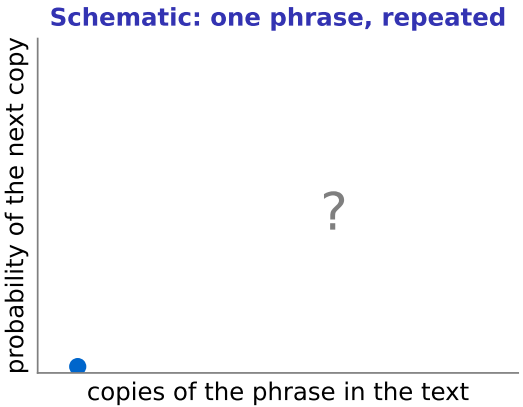
Under greedy decoding, the end of a text can be a loop of one phrase.

Probability of a Repeated Phrase

Under greedy decoding, the end of a text can be a loop: one phrase, repeated again and again.

Discovery

Each copy of the phrase is added to the context. With every copy, does the probability of the next copy grow or fall?



Under greedy decoding, the probability of a repeated phrase grows with each copy.

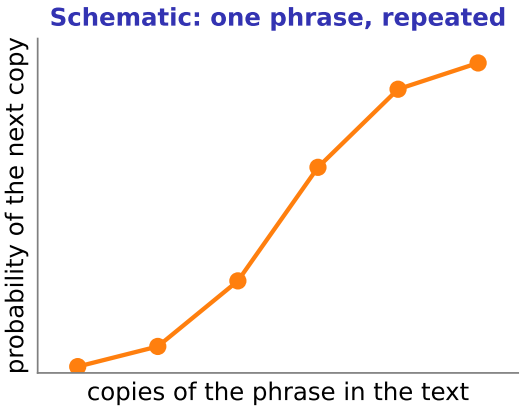
Probability of a Repeated Phrase

Under greedy decoding, the end of a text can be a loop: one phrase, repeated again and again.

Discovery

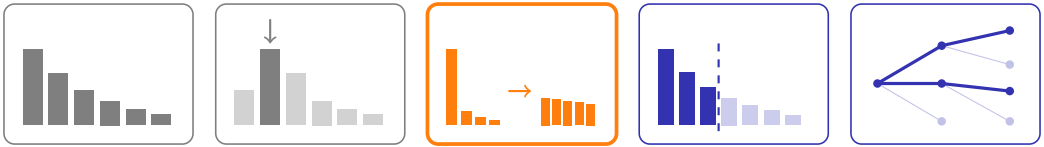
Each copy of the phrase is added to the context. With every copy, does the probability of the next copy grow or fall?

Answer: it grows. With each copy in the context, the next copy has a larger probability, and the loop is unbroken to the last token (Holtzman et al., 2020).



Under greedy decoding, the probability of a repeated phrase grows with each copy.

One Distribution, Five Decoding Rules



✓ finished ▶ next

One next-token distribution is the input of all five decoding rules.

Temperature

✓ 1 Next-Token Distribution

✓ 2 Greedy Decoding

▶ **3 Temperature**
Name the limits of $p^{(T)}$ as $T \to 0$, $T \to \infty$.

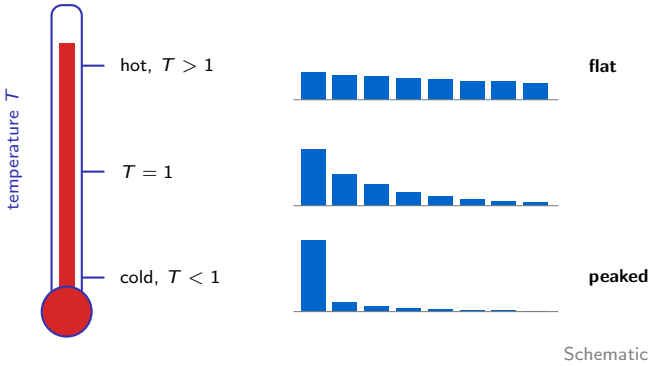
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Choose top-k or top-p for a peaked or a flat distribution.

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Run beam search of width B on a two-step tree.

6 Worked Example: GPT-2 Small
Compute each rule on the numbers of GPT-2 small after one prompt.

✓ finished ▶ next under each part ahead: its learning goal

Temperature From Peaked to Flat



A low temperature gives a peaked distribution, a high temperature a flat one.

Scores Divided by T Before the Softmax

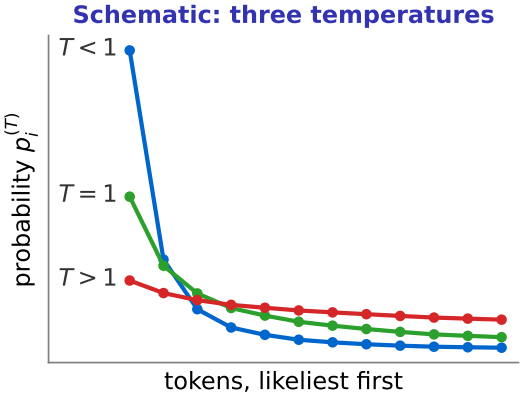
Definition

Under **sampling**, the next token is drawn at random from a distribution q. With **temperature** $T > 0$, $q = p^{(T)}$:

$$p_i^{(T)} = \frac{e^{z_i / T}}{\sum_{j \in V} e^{z_j / T}}.$$

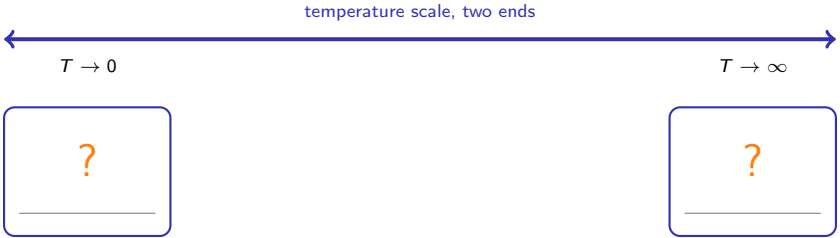
Intuition:

- $T < 1 \rightarrow$ more mass on the likeliest tokens
- $T > 1 \rightarrow$ the mass spread over more tokens
- The order of the tokens is the same at every T; $T = 1$ gives p



Under temperature T, the scores are divided by T before the softmax.

Two Ends of the Temperature Scale



At each end of the temperature scale, the distribution has a limit.

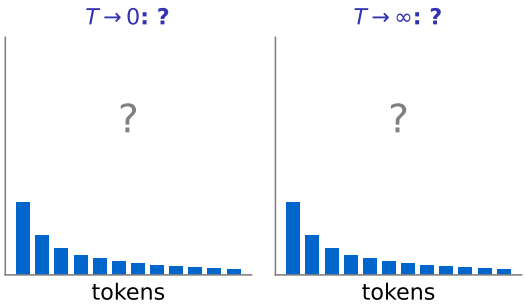
Two Limits of the Temperature

Temperature T in the softmax: $p_i^{(T)} \propto e^{z_i/T}$, with two limits.

Discovery

Which distribution is $p^{(T)}$ as $T \to 0$, which as $T \to \infty$?

Schematic: $p^{(T)}$ at $T = 1$



At $T \to 0$, sampling equals greedy decoding; at $T \to \infty$, every token has probability $1/|V|$.

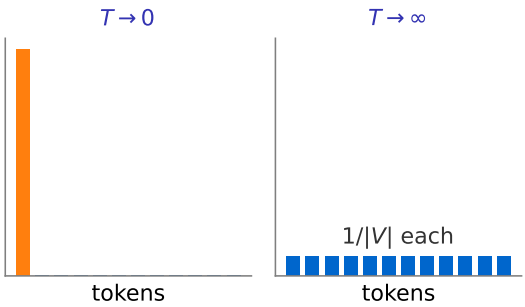
Two Limits of the Temperature

Temperature T in the softmax: $p_i^{(T)} \propto e^{z_i/T}$, with two limits.

Discovery

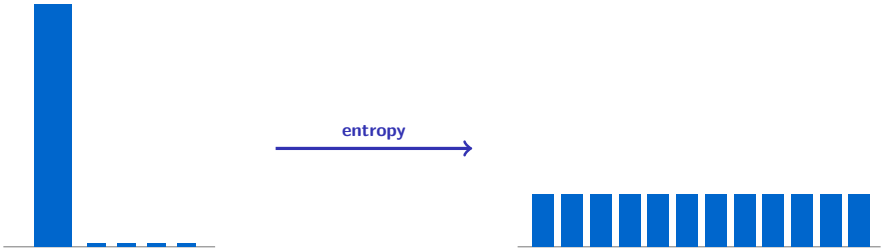
Which distribution is $p^{(T)}$ as $T \to 0$, which as $T \to \infty$?
Answer: $T \to 0$: all mass on the largest score, so a draw from $p^{(T)}$ equals greedy decoding. $T \to \infty$: $z_i/T \to 0$ for every token, so $p_i^{(T)} \to 1/|V|$, the uniform distribution.

Schematic: two limits of $p^{(T)}$



At $T \to 0$, sampling equals greedy decoding; at $T \to \infty$, every token has probability $1/|V|$.

Entropy From One Sure Token to All Tokens Equal



Schematic

The entropy is 0 when one token is sure, largest when all tokens are equal.

Entropy Grows With Temperature

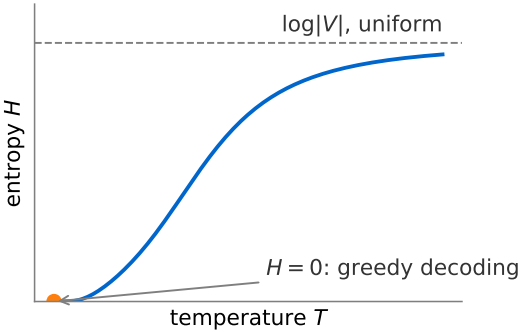
Definition

The **entropy** of a distribution p is $H(p) = -\sum_i p_i \log p_i$, in nats. $H = 0$ when one token has all the mass; $H = \log |V|$ under the uniform distribution.

Intuition:

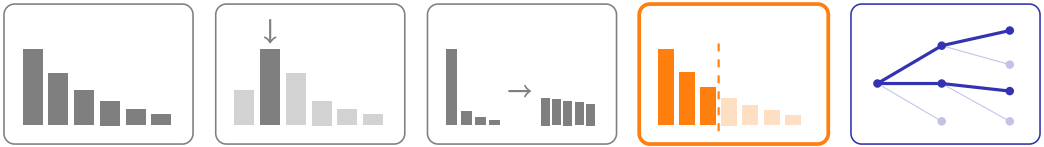
- $\frac{dH}{dT} = \frac{\mathrm{Var}_{p^{(T)}}(z)}{T^3} \geq 0$ (Appendix A2)
- H grows with T, from 0 to $\log |V|$
- A larger $H \rightarrow$ a less predictable next token

Schematic: entropy against T



The entropy of $p^{(T)}$ grows with the temperature, from 0 to $\log |V|$.

One Distribution, Five Decoding Rules



✓ finished ▶ next

One next-token distribution is the input of all five decoding rules.

Top-k, Top-p

✓ 1 Next-Token Distribution

✓ 2 Greedy Decoding

✓ 3 Temperature

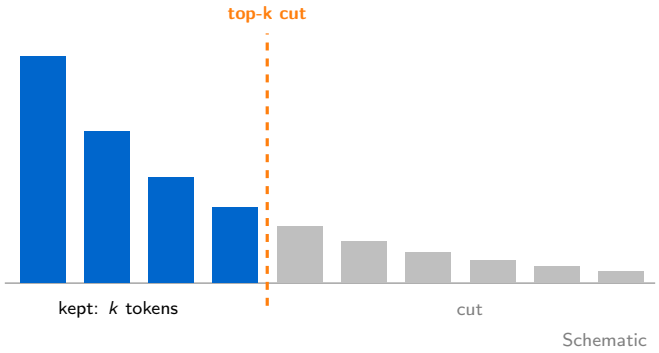
▶ **4 Top-k, Top-p**
Choose top-k or top-p for a peaked or a flat distribution.

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Run beam search of width B on a two-step tree.

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Compute each rule on the numbers of GPT-2 small after one prompt.

✓ finished ▶ next under each part ahead: its learning goal

Top-k: a Cut After k Tokens



Under top-k, every token after the k-th is cut.

Top-k: the k Likeliest Tokens

Definition

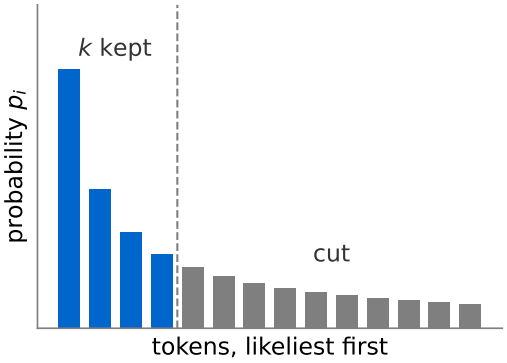
Under **top-k** sampling, the set S_k of the k likeliest tokens is kept, with the mass $M_k = \sum_{i \in S_k} p_i$. The draw is from

$$q_i = p_i / M_k \quad (i \in S_k), \qquad q_i = 0 \text{ otherwise.}$$

Intuition:

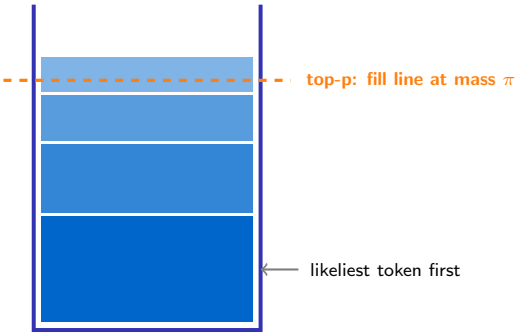
- The number of kept tokens is fixed at k
- The kept mass M_k depends on p
- $k = 1 \rightarrow$ greedy decoding; $k = |V| \rightarrow p$ itself

Schematic: top-k, kept tokens in blue



Under top-k, the k likeliest tokens are kept, whatever their mass.

Top-p: a Fill Line at Mass π





Schematic

Under top-p, the line π is fixed; the number of tokens up to it depends on p.

Top-p: the Smallest Set With Mass π

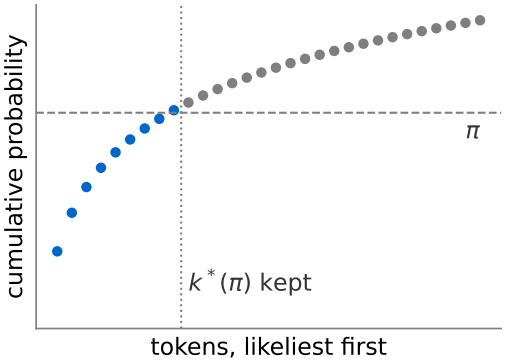
Definition

Under **top-p** sampling, or nucleus sampling, the kept set is the smallest set of likeliest tokens with mass $M \geq \pi$. The draw is from $q_i = p_i/M$ on that set; its size is $k^*(\pi)$.

Intuition:

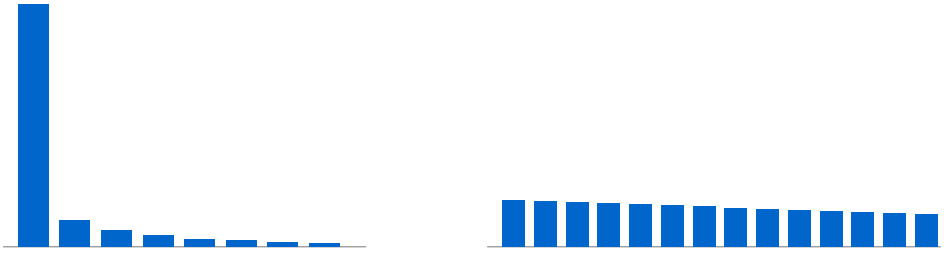
- The kept mass is fixed: $M \geq \pi$
- The number of kept tokens $k^*(\pi)$ depends on p
- $\pi \to 0 \to$ greedy decoding; $\pi = 1 \to p$ itself

Schematic: top-p, kept tokens in blue



Under top-p, the likeliest tokens are kept until their mass is at least π.

A Peaked, a Flat Distribution



Schematic

A peaked distribution has most of its mass on one token; a flat one has it on many tokens.

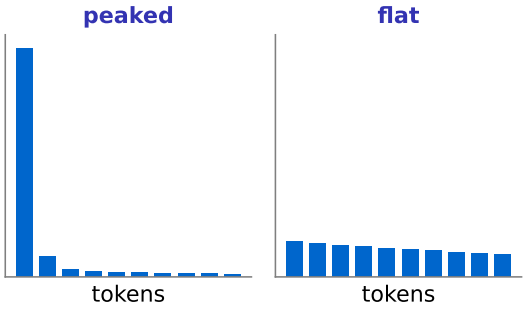
Fixed k Against Fixed π on Two Distributions

Two distributions: a peaked one, with most of the mass on one token; a flat one, with the mass spread over many tokens.

Discovery

One k, one π for both. Under top-k, after which distribution is the cut mass $1 - M_k$ larger? Under top-p, after which are more tokens kept?

Schematic: two distributions



Under top-k the number of kept tokens is fixed; under top-p, the kept mass.

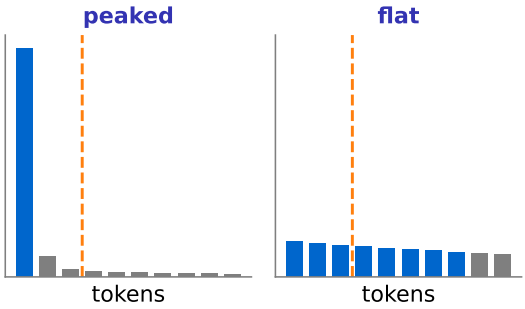
Fixed k Against Fixed π on Two Distributions

Two distributions: a peaked one, with most of the mass on one token; a flat one, with the mass spread over many tokens.

Discovery

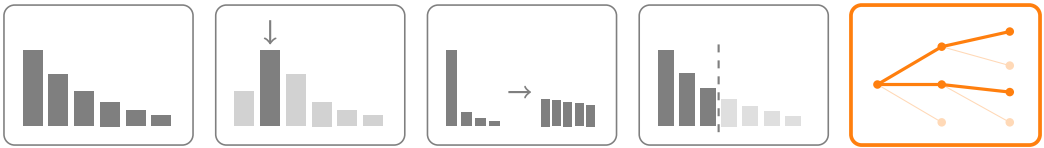
One k, one π for both. Under top-k, after which distribution is the cut mass $1 - M_k$ larger? Under top-p, after which are more tokens kept?
Answer: the flat one, both times. Under top-k, most of its mass is cut, and almost none of the peaked one. Under top-p, the kept mass is at least π after both, with many tokens of the flat one, few of the peaked one.

Schematic: dashed k; blue top-p set



Under top-k the number of kept tokens is fixed; under top-p, the kept mass.

One Distribution, Five Decoding Rules



One next-token distribution is the input of all five decoding rules.

Beam Search

✓ 1 Next-Token Distribution

✓ 2 Greedy Decoding

✓ 3 Temperature

✓ 4 Top-k, Top-p

▶ **5 Beam Search**

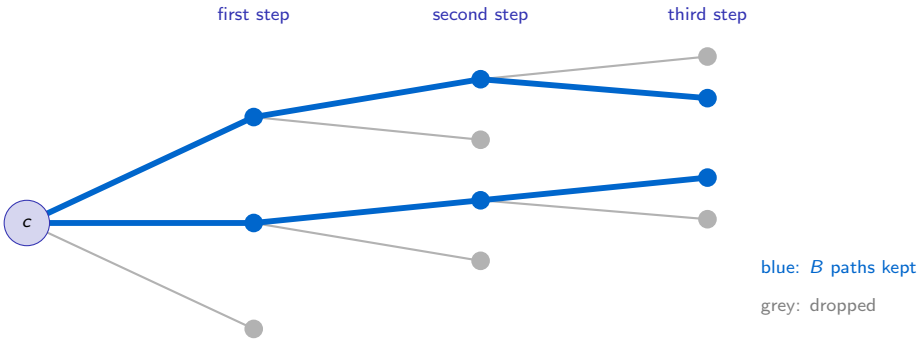
Run beam search of width B on a two-step tree.

6 Worked Example: GPT-2 Small

Compute each rule on the numbers of GPT-2 small after one prompt.

✓ finished ▶ next under each part ahead: its learning goal

Beam Search: B Paths Kept at Each Step



Under beam search, B paths are followed at every step; under greedy decoding, one.

Beam Search of Width B in Four Steps

Definition

Under **beam search** of width B, the B partial continuations with the largest $\log P$ are kept at every step; the result has the largest $\log P$ at the end.

The four steps:

1. **Start** with the context c as one partial continuation, $\log P = 0$
2. **Extend** each kept partial continuation by every token i: $\log P + \log p(i \mid \cdot)$
3. **Keep** the B largest of the $B \cdot |V|$ extensions
4. **Repeat** steps 2, 3 until n tokens are appended

Discovery

At $B = 1$, which decoding rule is beam search?

Beam Search of Width B in Four Steps

Definition

Under **beam search** of width B, the B partial continuations with the largest $\log P$ are kept at every step; the result has the largest $\log P$ at the end.

The four steps:

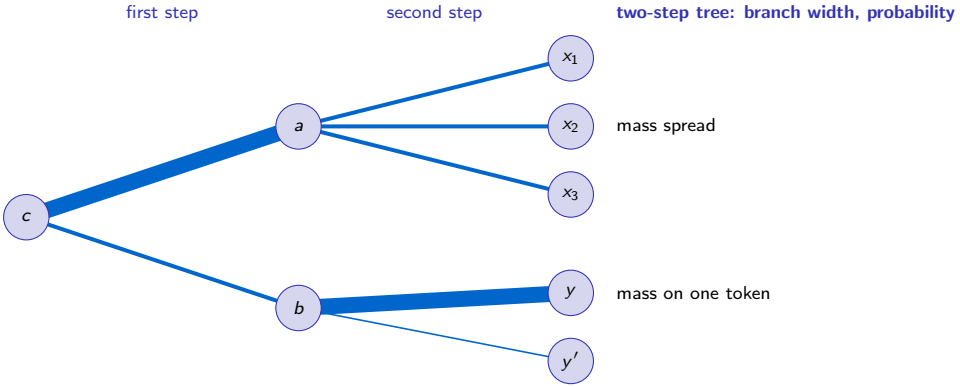
1. **Start** with the context c as one partial continuation, $\log P = 0$
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4. **Repeat** steps 2, 3 until n tokens are appended

Discovery

At $B = 1$, which decoding rule is beam search?
Greedy decoding: one partial continuation, extended by its likeliest token.

Beam search of width 1 is greedy decoding.

Branch Widths in a Two-Step Tree



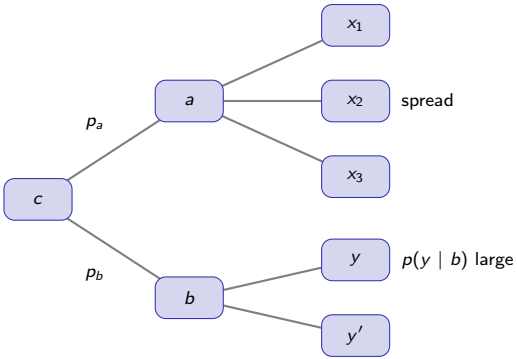
In the tree, the width of a branch is its probability.

Greedy Path Against the Most Probable Path

Two first tokens a, b with $p_a > p_b$. After a, the mass is spread over many tokens; after b, most of it is on one token y.

Discovery

Under which condition is the path $b\,y$ more probable than every path starting with a?



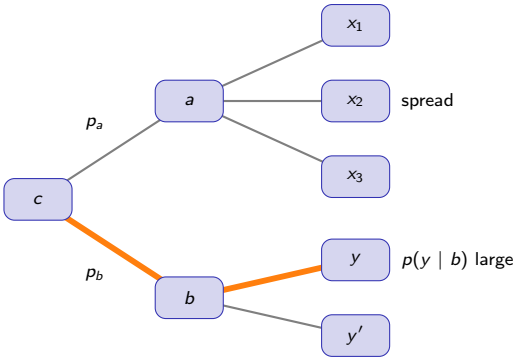
The first token of the most probable continuation can have the smaller probability.

Greedy Path Against the Most Probable Path

Two first tokens a, b with $p_a > p_b$. After a, the mass is spread over many tokens; after b, most of it is on one token y.

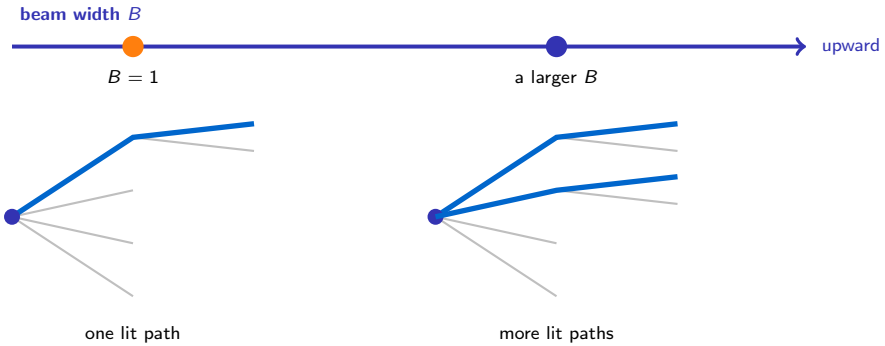
Discovery

Under which condition is the path $b\,y$ more probable than every path starting with a?
Answer: when $p_b\, p(y \mid b) > p_a \max_x p(x \mid a)$. Under greedy decoding, b is dropped at the first step. Under beam search with $B \geq 2$, b is kept, and $b\,y$ is the result.



The first token of the most probable continuation can have the smaller probability.

Beam Width B From 1 Upward



The beam width B is a whole number, at least 1.

Checkpoint: Beam Width, Search Result

Checkpoint problem

Beam search of width B over V, n steps after a context c.

Question 1: For $n = 2$, which smallest B gives the most probable two-token continuation after every context?

Checkpoint: Beam Width, Search Result

Checkpoint problem

Beam search of width B over V, n steps after a context c.

Question 1: For $n = 2$, which smallest B gives the most probable two-token continuation after every context?

Answer: $\mathbf{B} = |\mathbf{V}|$: every first token is kept, all $|V|^2$ paths are compared. With a smaller B, the best path can start with a dropped token.

Question 2: At the end, which is larger: $\log P$ at width B, or $\log P$ at width $B + 1$?

Checkpoint: Beam Width, Search Result

Checkpoint problem

Beam search of width B over V, n steps after a context c.

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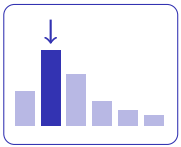
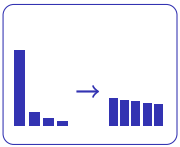
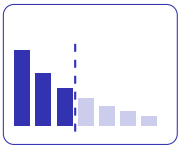
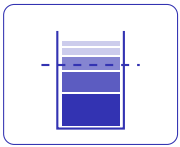
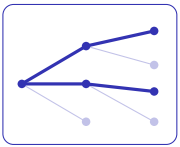
Answer: $\mathbf{B = |V|}$: every first token is kept, all $|V|^2$ paths are compared. With a smaller B, the best path can start with a dropped token.

Question 2: At the end, which is larger: $\log P$ at width B, or $\log P$ at width $B + 1$?

Answer: either one: at width $B + 1$, another set can be kept at an intermediate step, without the result of width B.

With a wider beam, more paths are compared, and the result can have a smaller $\log P$.

Five Rules, Four Parameters

five rules					
	greedy decoding	temperature	top-k	top-p	beam search
four parameters	**none**	T	k	π	B

Greedy decoding has no parameter; each other rule has one.

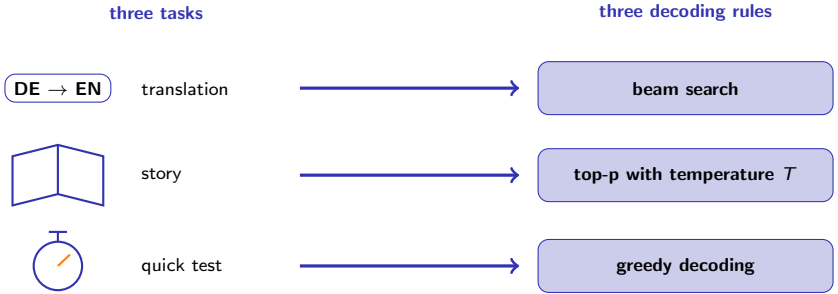
Five Decoding Rules in Six Aspects

Six aspects of the five decoding rules, at one step:

Aspect	Greedy	Temperature	Top-k	Top-p	Beam search
Kind	search	sampling	sampling	sampling	search
Parameter	none	T	k	π	B
One text per prompt	yes	no	no	no	yes
Kept at a step	1 token	all of V	k tokens	$k^*(\pi)$ tokens	B paths
Number kept depends on p	no	no	no	yes	no
Equal to greedy decoding at	every setting	$T \to 0$	$k = 1$	$\pi \to 0$	$B = 1$

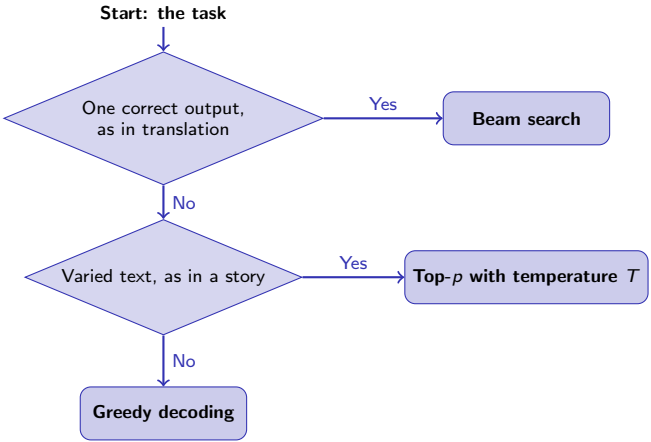
Greedy decoding is a limit of each of the four other rules.

Three Tasks, Three Decoding Rules



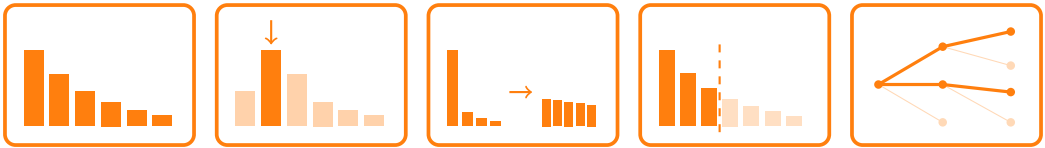
A translation, a story, a quick test: each task has its decoding rule.

From the Task to a Decoding Rule



The decoding rule is chosen from the task, before any text is generated.

One Distribution, Five Decoding Rules



each one on the numbers of GPT-2 small

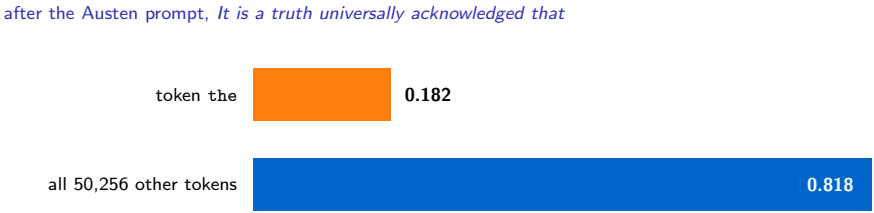
One next-token distribution is the input of all five decoding rules.

Worked Example: GPT-2 Small

- ✓ 1 Next-Token Distribution
- ✓ 2 Greedy Decoding
- ✓ 3 Temperature
- ✓ 4 Top-k, Top-p
- ✓ 5 Beam Search
- ▶ **6 Worked Example: GPT-2 Small**
  Compute each rule on the numbers of GPT-2 small after one prompt.

✓ finished ▶ next under each part ahead: its learning goal

The Token `the` Against 50,256 Other Tokens



After the Austen prompt, the likeliest token, `the`, has less than a fifth of the mass.

Ten Likeliest Tokens After the Austen Prompt

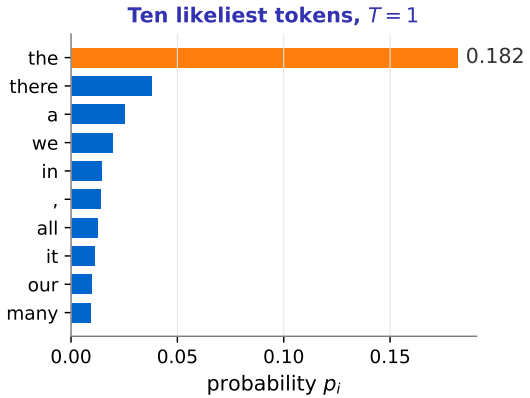
Definition

The **Austen prompt** is the context *It is a truth universally acknowledged that*, the first words of *Pride and Prejudice* (1813). GPT-2 small has $|V| = 50{,}257$ tokens.

Worked: the five likeliest next tokens at $T = 1$:

Token	Score z_i	p_i
`the`	−87.547	0.182
`there`	−89.120	0.038
`a`	−89.514	0.025
`we`	−89.788	0.019
`in`	−90.090	0.014
Five tokens		0.278

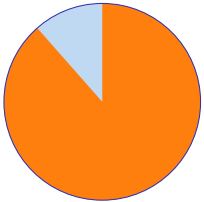
Worked: greedy decoding gives `the`; under top-p with $\pi = 0.9$, 3,784 tokens are kept.



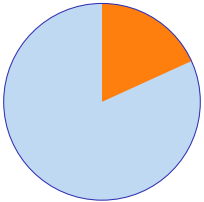
After the Austen prompt, the likeliest token has probability 0.182, and 3,784 tokens have a mass of 0.9.

Probability of the Token the at Three Temperatures

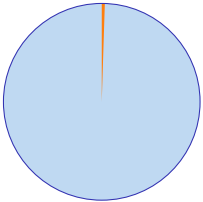
probability of the token the after the Austen prompt, three temperatures



$T = 0.5$: **0.885**



$T = 1$: **0.182**



$T = 2$: **0.005**

$p(\texttt{the})$ **is 0.885 at** $T = 0.5$**, 0.182 at** $T = 1$**, 0.005 at** $T = 2$**.**

Ratio of the Two Likeliest Tokens at T = 0.5, 1, 2

Step 1: the scores of the two likeliest tokens after the Austen prompt

Token	Score z_i
`the`	-87.547
`there`	-89.120
$\Delta = z_{\text{the}} - z_{\text{there}}$	1.573

Under temperature T, the ratio is $p_{\text{the}}^{(T)} / p_{\text{there}}^{(T)} = e^{\Delta/T}$.

Step 2: the ratio at three temperatures

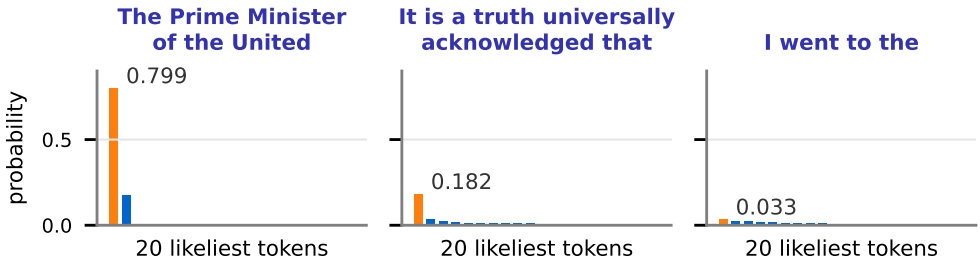
$$e^{1.573/0.5} = 23.2, \quad e^{1.573/1} = 4.82, \quad e^{1.573/2} = 2.20$$

Reading the results:

- $p_{\text{the}} = 0.885$ at $T = 0.5$, 0.182 at $T = 1$, 0.005 at $T = 2$
- $H = 0.69$, 6.38, 9.88 nats, against $\log |V| = 10.82$
- At every T, `the` is the likeliest token

At $T = 0.5$, `the` is 23.2 times as probable as `there`; at $T = 2$, 2.20 times.

Three Prompts, Three Shapes



The 20 likeliest tokens of GPT-2 small after three prompts: three shapes on one scale.

After the peaked prompt, one token has 0.799; after the flat prompt, the likeliest has 0.033.

Top-k Mass, Top-p Size After Three Prompts

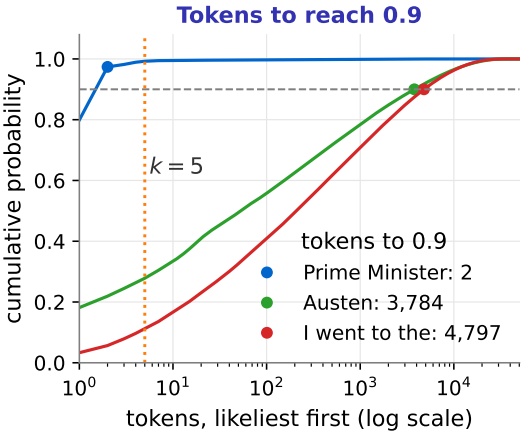
Definition

The **peaked prompt** is *The Prime Minister of the United*; the **flat prompt** is *I went to the*.

Worked: three prompts, top-k mass M_5, top-p size $k^*(0.9)$:

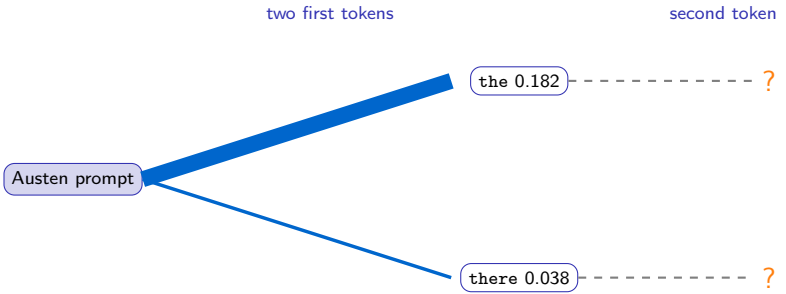
Prompt	$p_{\max}$	Mass M_5	Size $k^*(0.9)$
peaked	0.799	0.992	2
Austen	0.182	0.278	3,784
flat	0.033	0.112	4,797

Worked: under top-k with $k = 5$ after the flat prompt, `hospital` has $q = 0.033/0.112 = 0.295$; under top-p, 4,797 tokens are kept.



At $k = 5$, 0.992 of the mass is kept after the peaked prompt, 0.112 after the flat one.

Two First Tokens After the Austen Prompt



At the first step, `the` has 4.8 times the probability of `there`.

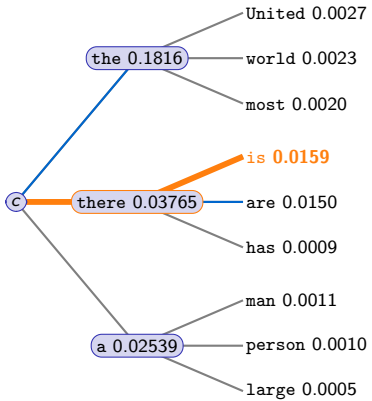
Nine Two-Token Paths After the Austen Prompt

Definition

After the Austen prompt, a **path** is a continuation of two tokens, with the probability $p(w_1 \mid c)\, p(w_2 \mid c, w_1)$. The tree has nine paths: three first tokens, three second tokens after each.

Path	$p(w_1)$	$p(w_2 \mid w_1)$	P
`the United`	0.1816	0.01472	0.0027
`there is`	0.03765	0.4223	0.0159
`there are`	0.03765	0.3976	0.0150

Worked: $B = 1$ gives `the United`, $0.1816 \times 0.01472 = 0.0027$; $B = 2$ gives `there is`, $0.03765 \times 0.4223 = 0.0159$, and $0.0159/0.0027 = 5.9$.



Leaves: P; blue: kept at $B = 2$; orange: the result.

With $B = 2$, the result is `there is`, 5.9 times as probable as the greedy path `the United`.

One Sentence Six Times

copy 1: *the United States is a nation of immigrants.*

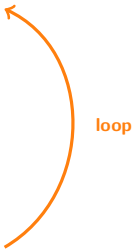
copy 2: *The United States is a nation of immigrants.*

copy 3: *The United States is a nation of immigrants.*

copy 4: *The United States is a nation of immigrants.*

copy 5: *The United States is a nation of immigrants.*

copy 6: *The United States is a nation of immigrants.*



Under greedy decoding after the Austen prompt, one sentence of GPT-2 small is repeated six times.

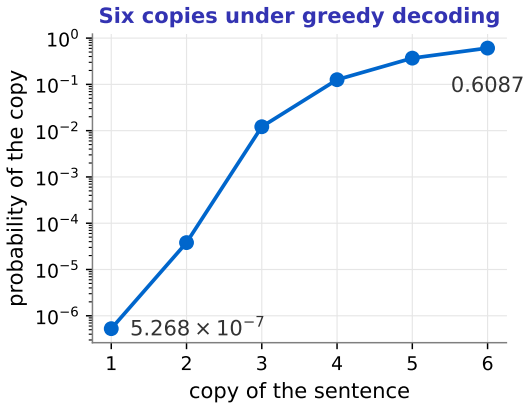
Six Copies of One Sentence Under Greedy Decoding

Definition

A **copy** is one occurrence of the sentence *The United States is a nation of immigrants.* in the greedy continuation of the Austen prompt. Its probability P is the product of its 9 token probabilities.

Copy	Tokens	P
1	1 to 9	5.268×10^{-7}
2	10 to 18	3.812×10^{-5}
3	19 to 27	0.01216
4	28 to 36	0.1263
5	37 to 45	0.3692
6	46 to 54	0.6087

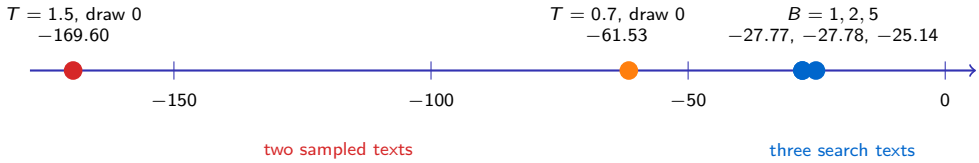
Worked: from copy 1 to copy 6, P grows by $0.6087/(5.268 \times 10^{-7}) = 1.16 \times 10^{6}$.



Under greedy decoding, the probability of the sentence grows with every copy, from 5.268×10^{-7} to 0.6087.

Five Values of log P on One Line

five values of log P on one number line, 20 tokens after the Austen prompt



The three search texts lie near -27; the two sampled texts lie far below.

Twenty Tokens After the Austen Prompt Under Five Rules

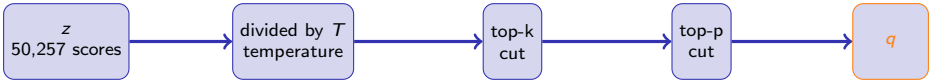
Five rules, 20 tokens each after the Austen prompt; $\log P$ is the sum of the 20 log-probabilities of GPT-2 small at $T = 1$.

Rule	$\log P$	Continuation
greedy, $B = 1$	-27.77	the United States is a nation of immigrants. The United States is a nation of immigrants. The United
beam, $B = 2$	-27.78	there is no such thing as a ‘good’ or ‘bad’ person. ↩↩ It is
beam, $B = 5$	-25.14	there is no such thing as a “right” or “wrong” way of life. It is
$T = 0.7$, draw 0	-61.53	the best and the brightest minds in the world face incredible pain and surmounting adversity. The best
$T = 1.5$, draw 0	-169.60	WWI and the 1914 Ottoman Imperial Question originally parted rows) Leatherwolves moreapa hamley ancient

Over 20 tokens, $\log P$ is -25.14 under beam search of width 5 and -169.60 for a draw at $T = 1.5$.

Scores Through Temperature, Top-k, Top-p

`next_token_dist(z, T, k, top_p)`



In `next_token_dist`, the scores are divided by T first, then cut by top-k, top-p.

next_token_dist on the Flat Prompt

```
z = logits("I went to the")                  # 50,257 scores
p = next_token_dist(z, T=1.0)                # T = 1, no draw
q = next_token_dist(z, T=1.0, k=5)           # top-k
r = next_token_dist(z, T=1.0, top_p=0.9)     # top-p
print(word(p.argmax()), round(p.max(), 4))   # hospital 0.0331
print((q > 0).sum(), round(q.max(), 4))      # 5 0.295
print((r > 0).sum())                         # 4797
```

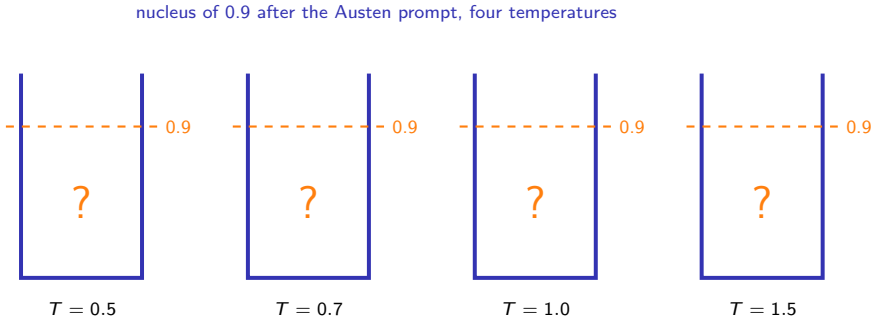
Key outputs:

- `p.max()`: 0.0331, the probability of `hospital`
- `q`: 5 tokens kept, `hospital` at 0.295
- `r`: 4,797 tokens kept at $\pi = 0.9$

Note: $0.0331/0.1122 = 0.295$, the value of top-5 after the flat prompt.

Temperature, top-k, top-p are arguments of one function, `next_token_dist`.

Nucleus at Four Temperatures



After the Austen prompt, the size of the nucleus of 0.9 depends on T.

Nucleus Size at Four Temperatures

Discovery Exercise

Open **notebook 01** in Colab, run the setup cell, then the four lines, at four temperatures:

```
z = logits(AUSTEN)
for T in [0.5, 0.7, 1.0, 1.5]:           # four distributions, no draw
    r = next_token_dist(z, T=T, top_p=0.9)
    print(T, (r > 0).sum(), round(r.max(), 3))
```

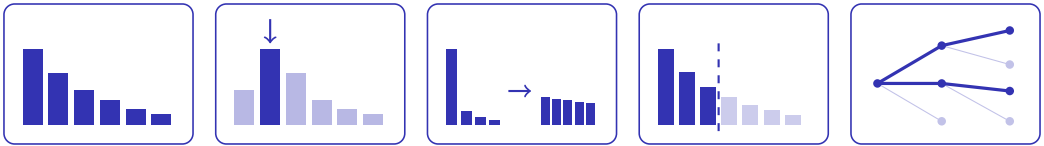
Questions to answer:

1. What is the size of the nucleus of 0.9 at each T?
2. What is the probability of `the` after the cut at $T = 0.5$, against 0.885 before it?
3. In **notebook 03**, what is the smallest width B with the result `there is`?

Derivations with checks on GPT-2 small: **notebook 02**.

After the Austen prompt, the nucleus of 0.9 grows with the temperature.

One Distribution, Five Decoding Rules



✓ 1 Next-Token Distribution | ✓ 2 Greedy Decoding | ✓ 3 Temperature | ✓ 4 Top-k, Top-p | ✓ 5 Beam Search

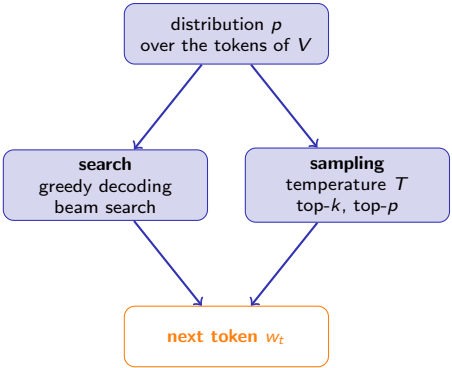
✓ finished

One next-token distribution is the input of all five decoding rules.

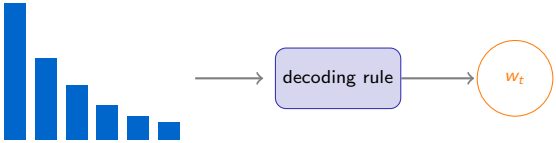
Five Results on Decoding the Next Token

Five results on decoding:

1. Softmax: $p_i = e^{z_i} / \sum_j e^{z_j}$; P of a continuation is a product
2. Greedy decoding: $\arg\max_i p_i$; the probability of a repeated phrase grows with each copy
3. $p^{(T)} \propto e^{z/T}$: greedy decoding at $T \to 0$, uniform at $T \to \infty$
4. Top-k: k tokens fixed; top-p: the mass π fixed
5. Beam search: B paths kept, $B = 1$ greedy; with a wider beam, $\log P$ can be smaller



Each of the five rules is a map from one distribution per step to the next token.



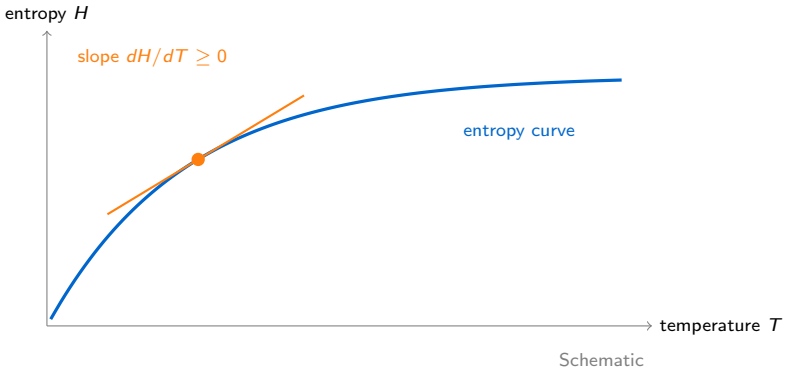
Thank you

Picking the Next Token
Text Analytics, Chapter 7

Appendix

Derivations: entropy, truncation, min-p, repetition penalty

Slope of the Entropy Curve



Schematic

The slope of the entropy in T is at least zero at every T.

A2: Derivative of the Entropy in T

Given $p_i^{(T)} = e^{\beta z_i}/Z(\beta)$ with $\beta = 1/T$, $Z(\beta) = \sum_j e^{\beta z_j}$; expectations under $p^{(T)}$.

Step 1: $\log p_i = \beta z_i - \log Z$, so the entropy is

$$H = -\sum_i p_i \log p_i = \log Z - \beta\, \mathbb{E}[z].$$

Step 2: $\dfrac{d \log Z}{d\beta} = \mathbb{E}[z]$ and $\dfrac{d\, \mathbb{E}[z]}{d\beta} = \mathrm{Var}(z)$, so

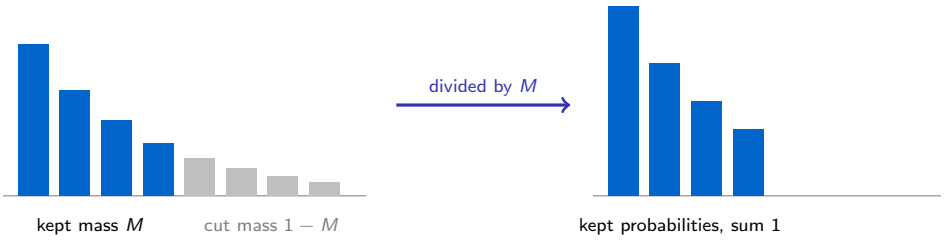
$$\frac{dH}{d\beta} = \mathbb{E}[z] - \mathbb{E}[z] - \beta\, \mathrm{Var}(z) = -\beta\, \mathrm{Var}(z).$$

Step 3: with $d\beta/dT = -1/T^2$, the derivative in T is

$$\frac{dH}{dT} = \frac{\mathrm{Var}_{p^{(T)}}(z)}{T^3} \;\geq\; 0.$$

The entropy of $p^{(T)}$ grows with T at the rate $\mathrm{Var}(z)/T^3$.

Kept Mass M, Cut Mass $1 - M$



Schematic

After a cut, the kept probabilities are divided by the kept mass M.

A3: Divergence of a Truncation

Given a kept set S with mass $M = \sum_{i \in S} p_i$; $q_i = p_i / M$ on S, $q_i = 0$ elsewhere. Top-k, top-p, min-p are truncations of this form.

Step 1: on S, $q_i / p_i = 1/M$, so

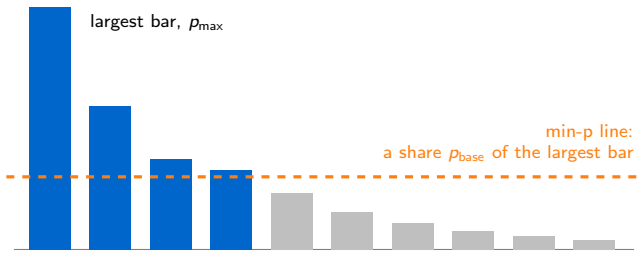
$$\mathrm{KL}(q \,\|\, p) = \sum_{i \in S} q_i \log \frac{q_i}{p_i} = \log \frac{1}{M} \sum_{i \in S} q_i = -\log M.$$

Step 2: under top-p, $M \geq \pi$, so $\mathrm{KL}(q \,\|\, p) \leq -\log \pi$ after every prompt.

Step 3: under top-k, the k largest of $|V|$ probabilities have a mean of at least $1/|V|$, so $M_k \geq k/|V|$ and $\mathrm{KL}(q \,\|\, p) \leq \log(|V|/k)$.

The divergence of a truncation is $-\log M$; it depends on the kept mass alone.

Min-p: a Line at a Share of the Largest Bar



Under min-p, a token is kept when its probability is at least a share of the largest one.

A4: Bound on the Number of Tokens Under Min-p

Definition

Under **min-p** (Nguyen et al., 2024), the kept set is $S = \{\, i : p_i \geq p_{\text{base}}\, p_{\max} \,\}$, with $0 < p_{\text{base}} < 1$.

Step 1: each kept token other than the likeliest has $p_i \geq p_{\text{base}}\, p_{\max}$; together these tokens have a mass of at most $1 - p_{\max}$.

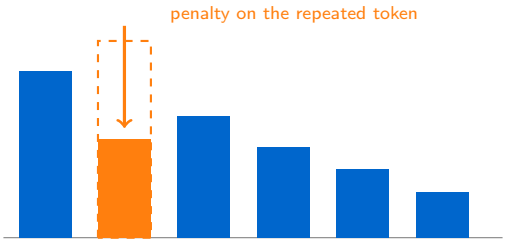
Step 2:

$$(|S| - 1)\, p_{\text{base}}\, p_{\max} \leq 1 - p_{\max} \;\Rightarrow\; |S| \leq 1 + \frac{1 - p_{\max}}{p_{\text{base}}\, p_{\max}}.$$

Interpretation: a peaked p, with $p_{\max}$ near $1 \to$ few tokens; a flat $p \to$ the bound grows as $1/p_{\max}$.

Under min-p, the number of kept tokens is at most $1 + (1 - p_{\max})/(p_{\text{base}}\, p_{\max})$.

Penalty on a Repeated Token



Schematic

Under the repetition penalty, the score of a token already in the text is lowered.

A5: Repetition Penalty, Offset of the Scores

Definition

Under the **repetition penalty** $\theta > 1$ (Keskar et al., 2019), the score of each token already in the text is changed to

$$z_i' = \begin{cases} z_i/\theta, & z_i > 0, \\ \theta\, z_i, & z_i \leq 0. \end{cases}$$

Step 1: $\mathrm{softmax}(z + a) = \mathrm{softmax}(z)$ for every constant a: p is the same at every offset of the scores.

Step 2: the penalty depends on the offset: for $z_i < 0$, the score is lower by $(\theta - 1)\,|z_i|$, a difference that grows with $|z_i|$.

Interpretation: two score vectors with one softmax p can give two different distributions after the penalty.

The repetition penalty depends on the offset of the scores; the softmax is the same at every offset.